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STORAGE MEDIUM STORING AN APPLICATION PROGRAM, CONTROL METHOD FOR INFORMATION PROCESSING APPARATUS, AND INFORMATION PROCESSING APPARATUS

Abstract

A non-transitory computer-readable storage medium stores an application program for using a service that publishes post information for causing an information processing apparatus comprising the application program to execute a determining step of determining whether or not an image included in accepted post information is an image generated by a generative AI, and a restricting step of, in a case of being determined that the image included in the accepted post information is an image generated by the generative AI, restricting posting of the accepted post information.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to a storage medium that stores an application program, a control method for an information processing apparatus, and an information processing apparatus.

Description of the Related Art

[0002] In recent years, image generation technology using artificial intelligence (AI) has become widespread, and even individuals are able to easily generate images by using generative AI. This makes it possible for anyone to easily generate realistic images, but on the other hand, there have been occasional cases of malicious users using the generative AI to generate fabricated images of fictitious incidents or fictitious accidents, which has become a problem.

[0003] Incidentally, the use of social networking services (SNS), which allow users to post messages and images and interact with each other via the Internet, is increasing. As the use of the SNS increases, there is a problem of inappropriate content being posted. In order to deal with such a problem, techniques have been proposed for analyzing post information and detecting post information that includes inappropriate content. For example, in a technique disclosed in Japanese Laid-Open Patent Publication (kokai) No. 2022-39764, feature quantities are extracted from character string(s), image(s), and audio that are included in post information, and compared with information in a database to determine whether or not the post information includes inappropriate content.

[0004] However, with the technique disclosed in Japanese Laid-Open Patent Publication (kokai) No. 2022-39764, even in the case that the image included in the post information is a fabricated image generated by generative AI, if the image included in the post information is an image including information that could actually occur, the image included in the post information may not be determined to be inappropriate. In such a case, there is a concern that the post information including an AI-generated image that has fabricated a fictitious incident or accident has been made public, causing viewers to misidentify the post information.

SUMMARY OF THE INVENTION

[0005] The present invention provides a storage medium that stores an application program, a control method for an information processing apparatus, and an information processing apparatus.

[0006] Accordingly, the present invention provides a non-transitory computer-readable storage medium storing an application program for using a service that publishes post information for causing an information processing apparatus comprising the application program to execute a determining step of determining whether or not an image included in accepted post information is an image generated by a generative AI, and a restricting step of, in a case of being determined that the image included in the accepted post information is an image generated by the generative AI, restricting posting of the accepted post information.

[0007] Accordingly, the present invention provides a non-transitory computer-readable storage medium storing an application program for using a service that publishes post information for causing an information processing apparatus comprising the application program to execute a determining step of determining whether or not an image included in accepted post information is an image generated by a generative AI, and a step of, in a case of being determined that the image included in the accepted post information is an image generated by the generative AI, performing a modification processing for preventing misidentification of viewers on the accepted post information.

[0008] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a block diagram that schematically shows a configuration of a communication system that includes a communication apparatus as an information processing apparatus according to an embodiment of the present invention.

[0010] FIG. 2 is a block diagram that schematically shows a configuration of an AI image generation server shown in FIG. 1.

[0011] FIG. 3 is a block diagram that schematically shows a configuration of the communication apparatus shown in FIG. 1.

[0012] FIG. 4 is a block diagram that schematically shows a configuration of an SNS server shown in FIG. 1.

[0013] FIG. 5A and FIG. 5B are flowcharts that show the steps of an image generation processing by a generative AI in the communication system shown in FIG. 1.

[0014] FIG. 6A and FIG. 6B are flowcharts that show the steps of a first posting control processing executed by the communication system shown in FIG. 1.

[0015] FIGS. 7A, 7B and 7C are diagrams that show examples of screens displayed on a user I/F unit shown in FIG. 3.

[0016] FIG. 8A and FIG. 8B are flowcharts that show the steps of a second posting control processing executed by the communication system shown in FIG. 1.

[0017] FIG. 9A and FIG. 9B are diagrams for explaining a composite image generated by the process of S814 in FIG. 8B.

[0018] FIG. 10A and FIG. 10B are flowcharts that show the steps of a third posting control processing executed by the communication system shown in FIG. 1.

[0019] FIG. 11 is a diagram that shows an example of post information modified by the process of S1014 in FIG. 10B.

[0020] FIG. 12A and FIG. 12B are flowcharts that show the steps of a fourth posting control processing executed by the communication system shown in FIG. 1.

[0021] FIG. 13 is a diagram that shows an example of post information modified by the process of S1214 in FIG. 12B.

[0022] FIG. 14A and FIG. 14B are flowcharts that show the steps of a fifth posting control processing executed by the communication system shown in FIG. 1.

[0023] FIG. 15 is a flowchart that shows the procedure of a post information transmission processing executed by the communication apparatus shown in FIG. 1.

[0024] FIG. 16 is a flowchart that shows another procedure of the post information transmission processing executed by the communication apparatus shown in FIG. 1.

DESCRIPTION OF THE EMBODIMENTS

[0025] The present invention will now be described in detail below with reference to the accompanying drawings showing embodiments thereof.

[0026] Hereinafter, an embodiment of the present invention will be described in detail with reference to the accompanying drawings. It should be noted that the following embodiment does not limit the present invention as defined by the claims, and not all of the combinations of features described in the present embodiment are necessarily essential to the solving means of the present invention.

[0027] FIG. 1 is a block diagram that schematically shows a configuration of a communication system 100 that includes a communication apparatus 102 as an information processing apparatus

according to the embodiment of the present invention.

[0028] As shown in FIG. 1, the communication system **100** includes an AI image generation server **101**, the communication apparatus **102**, and an SNS server **103**. The AI image generation server **101**, the communication apparatus **102**, and the SNS server **103** are able to communicate with each other via a network **104**. In the communication system **100**, the communication apparatus **102** makes a generation request to generate an image according to an instruction received from a user. The AI image generation server **101** generates an image (hereinafter, referred to as “an AI-generated image”) by using a generative AI in accordance with the generation request received from the communication apparatus **102**, and transmits the generated AI-generated image to the communication apparatus **102**. The user operates the communication apparatus **102** to create post information. This post information includes, for example, the AI-generated image that has been generated by the AI image generation server **101**. The communication apparatus **102** transmits, to the SNS server **103**, the post information that has been created by the user. The SNS server **103** is a server that provides an SNS service for publishing post information. The SNS server **103** registers the received post information in a post information database **405** shown in FIG. 4, which will be described below. As a result, a plurality of users who use the SNS service are able to view this post information.

[0029] FIG. 2 is a block diagram that schematically shows a configuration of the AI image generation server **101** shown in FIG. 1. As shown in FIG. 2, the AI image generation server **101** includes a central processing unit (a CPU) **201**, a random access memory (a RAM) **202**, a read only memory (a ROM) **203**, a storage unit **204**, a graphics processing unit (a GPU) **207**, an AI-generated identification information adding unit **208**, and a network interface unit (a network I/F unit) **209**. The CPU **201**, the RAM **202**, the ROM **203**, the storage unit **204**, the GPU **207**, the AI-generated identification information adding unit **208**, and the network I/F unit **209** are connected to each other via a bus **210**.

[0030] The CPU **201** controls the operations of the AI image generation server **101** and runs based on a program that has been stored in the RAM **202**. The ROM **203** is a boot ROM, and stores a boot program and the like of the system. The storage unit **204** is a nonvolatile device such as a hard disk drive (an HDD) or a solid state drive (an SSD), and stores a trained model **205** and an AI image generation program **206** that are used to realize the generative AI. It should be noted that, as an image generation means of the AI image generation server **101**, a configuration will be described in which an arbitrary trained model **205** and an existing AI image generation program **206** such as Stable Diffusion are applied, but other image generation means may also be used. The trained model **205** and the AI image generation program **206** are loaded into the RAM **202** and are executed by the CPU **201**. In the AI image generation server **101**, when a generation request is received from the communication apparatus **102** or the like, the GPU **207** executes an AI image generation processing that generates an AI-generated image in accordance with instructions from the CPU **201**.

[0031] The AI-generated identification information adding unit **208** adds, to the AI-generated image that has been generated by the AI image generation processing, AI-generated identification information for identifying that it is an image generated by the generative AI. This AI-generated identification information includes information indicating that it is an image generated by the generative AI, information regarding the trained model **205** and the AI image generation program **206** that have been used, information included in the generation request, etc. It should be noted that the format of the AI-generated identification information may be metadata such as header information of an image file, or the AI-generated identification information may be superimposed on the image itself as an invisible digital watermark or a visible text. The AI-generated image, to which the AI-generated identification information has been added, is transmitted to the communication apparatus **102** via the network **104** or is stored in the storage unit **204**.

[0032] The network I/F unit **209** is connected to the network **104**, and performs inputting and

outputting various types of information via the network **104**.

[0033] FIG. **3** is a block diagram that schematically shows a configuration of the communication apparatus **102** shown in FIG. **1**. The communication apparatus **102** is an apparatus capable of communicating with the AI image generation server **101** and the SNS server **103**, such as a smartphone, a tablet terminal, a personal computer (a PC), or the like.

[0034] As shown in FIG. **3**, the communication apparatus **102** includes a CPU **301**, a RAM **302**, an SSD **303**, a user I/F unit **304**, and a network I/F unit **305**. The CPU **301**, the RAM **302**, the SSD **303**, the user I/F unit **304**, and the network I/F unit **305** are connected to each other via a bus **306**.

[0035] The CPU **301** controls the operations of the communication apparatus **102** and runs based on a program that has been loaded into the RAM **302**. The SSD **303** stores system programs and the like. In addition, the SSD **303** stores a program for an SNS application (a program of an SNS application). The SNS application is an application for using the SNS service provided by the SNS server **103**. The user I/F unit **304** includes a display, a touch panel, a keyboard, a mouse, etc., and performs input/output processing with the user. The network I/F unit **305** may be configured to be able to perform wired communication, or may be configured to be able to perform wireless communication. It should be noted that in the present embodiment, the communication apparatus **102** may include, in addition to the components shown in FIG. **3**, a speech unit for implementing a telephone function, a photographing unit for implementing a camera function, and the like.

[0036] FIG. **4** is a block diagram that schematically shows a configuration of the SNS server **103** shown in FIG. **1**. As shown in FIG. **4**, the SNS server **103** includes a CPU **401**, a RAM **402**, a ROM **403**, a storage unit **404**, an AI-generated identification information analyzing unit **406**, and a network I/F unit **407**. The CPU **401**, the RAM **402**, the ROM **403**, the storage unit **404**, the AI-generated identification information analyzing unit **406**, and the network I/F unit **407** are connected to each other via a bus **408**.

[0037] The CPU **401** controls the operations of the SNS server **103** and runs based on a program that has been stored in the RAM **402**. The ROM **403** is a boot ROM, and stores a boot program and the like of the system. The storage unit **404** is a storage unit that uses a nonvolatile device, such as an SSD, an HDD, or an embedded multi-media card (an eMMC). The storage unit **404** stores the post information database **405** for managing the post information received from an apparatus such as the communication apparatus **102**. The AI-generated identification information analyzing unit **406** analyzes the image to determine the presence or absence of the AI-generated identification information that has been added by the AI-generated identification information adding unit **208**. The network I/F unit **407** is connected to the network **104**, and performs inputting and outputting various types of information via the network **104**.

[0038] Next, a processing of generating an image by the generative AI in the communication system **100** will be described.

[0039] FIG. **5A** and FIG. **5B** are flowcharts that show the steps of an image generation processing by the generative AI in the communication system **100** shown in FIG. **1**. This image generation processing is performed by the communication apparatus **102** and the AI image generation server **101**.

[0040] FIG. **5A** is a flowchart that shows the steps of a generation request transmission processing performed by the communication apparatus **102**. The generation request transmission processing is realized by the CPU **301** of the communication apparatus **102** executing a program that has been stored in the SSD **303** or the like.

[0041] As shown in FIG. **5A**, first, the CPU **301** accepts image generation parameters for generating an AI-generated image from the user (**S501**). The image generation parameters are those used in publically-known AI image generation technologies, such as keywords, sentences (prompts), images, etc. related to the image to be generated. Next, the CPU **301** generates a generation request to generate an AI-generated image based on the accepted image generation parameters, and transmits this generation request to the AI image generation server **101** (**S502**).

After that, the generation request transmission processing ends.

[0042] FIG. 5B is a flowchart that shows the steps of an image generation control processing performed by the AI image generation server **101**. The image generation control processing is realized by the CPU **201** of the AI image generation server **101** executing a program that has been stored in the ROM **203** or the storage unit **204**.

[0043] As shown in FIG. 5B, when the CPU **201** receives the generation request from the communication apparatus **102** via the network **104** (S503), the CPU **201** controls the GPU **207** to perform the AI image generation processing (S504). As a result, an AI-generated image, which corresponds to the image generation parameters that have been inputted by the user, is generated. Next, the CPU **201** controls the AI-generated identification information adding unit **208** to add, to the AI-generated image that has been generated in S504, AI-generated identification information indicating that it is an image generated by the generative AI (S505). Next, the CPU **201** transmits the AI-generated image, to which the AI-generated identification information has been added, to the communication apparatus **102** via the network **104** (S506). After that, the image generation control processing ends.

[0044] Next, a processing related to posting of post information in the communication system **100** will be described.

[0045] FIG. 6A and FIG. 6B are flowcharts that show the steps of a first posting control processing executed by the communication system **100** shown in FIG. 1. The first posting control processing is performed by the communication apparatus **102** and the SNS server **103**.

[0046] FIG. 6A is a flowchart that shows the steps of a post information transmission processing performed by the communication apparatus **102**. The post information transmission processing is realized by the CPU **301** of the communication apparatus **102** executing the program for the SNS application (the program of the SNS application) that has been stored in the SSD **303**.

[0047] As shown in FIG. 6A, the CPU **301** accepts the post information from the user (S601). The post information includes at least any one of text data, audio data, image data, and moving image data (video data). In the present embodiment, the CPU **301** accepts the post information on a posting screen **701** that is shown in FIG. 7A and is displayed on the user I/F unit **304** (a display unit) of the communication apparatus **102**. The posting screen **701** includes a character input section **702**, a file selection section **703**, and a post button **704**. When the user selects one file by the file selection section **703**, information about the selected file is displayed on the posting screen **701**. As an example, in FIG. 7A, an image **705** that has been selected by the user in the file selection section **703** is displayed on the posting screen **701**. When the user presses the post button **704**, the CPU **301** accepts the post information.

[0048] Next, the CPU **301** transmits the accepted post information to the SNS server **103** (S602). The SNS server **103** performs a post information registration processing shown in FIG. 6B, which will be described below, and, for example, in the case that the received post information includes an AI-generated image, the

[0049] SNS server **103** transmits, to the communication apparatus **102**, a warning notification indicating that the received post information includes an AI-generated image.

[0050] Next, the CPU **301** determines whether or not the warning notification has been received from the SNS server **103** (S603).

[0051] In the case of being determined in S603 that the warning notification has not been received from the SNS server **103**, the post information transmission processing ends. On the other hand, in the case of being determined in S603 that the warning notification has been received from the SNS server **103**, the CPU **301** performs a processing of restricting posting of the post information accepted in S601. Specifically, the CPU **301** causes a warning notification screen **706** of FIG. 7B to be displayed on the posting screen **701** displayed on the user I/F unit **304** (S604). The warning notification screen **706** is a screen for allowing the user (a poster) to select whether or not to post the post information accepted in S601 as is, and includes a post button **707** and a cancel button **708**.

[0052] Next, the CPU **301** notifies the SNS server **103** of information about the button pressed by the user, either the post button **707** or the cancel button **708** (**S605**), and the post information transmission processing ends.

[0053] FIG. **6B** is a flowchart that shows the steps of the post information registration processing performed by the SNS server **103**. The post information registration processing is realized by the CPU **401** of the SNS server **103** executing a program that has been stored in the ROM **403** or the storage unit **404**.

[0054] As shown in FIG. **6B**, the CPU **401** receives the post information that has been transmitted from the communication apparatus **102** (**S611**). Next, the CPU **401** analyzes the received post information and determines whether or not this post information includes an image (**S612**).

[0055] In the case of being determined in **S612** that this post information includes an image, the CPU **401** controls the AI-generated identification information analyzing unit **406** to determine whether or not the image included in the post information is an AI-generated image (**S613**). For example, in the case that AI-generated identification information has been added to the image included in the post information, it is determined that the image included in the post information is an AI-generated image. On the other hand, in the case that AI-generated identification information has not been added to the image included in the post information, it is determined that the image included in the post information is not an AI-generated image.

[0056] In the case of being determined in **S613** that the image included in the post information is an AI-generated image, the CPU **401** transmits, to the communication apparatus **102**, a warning notification indicating that the post information received in **S611** includes an AI-generated image (**S614**). In the communication apparatus **102** that has received this warning notification, the warning notification screen **706** is displayed in **S604** described above. Furthermore, in **S605** described above, the information about the button pressed by the user, either the post button **707** or the cancel button **708** is notified to the SNS server **103**.

[0057] When the CPU **401** receives the information about the button pressed by the user from the communication apparatus **102** (**S615**), the CPU **401** determines whether the button pressed by the user is the post button **707** or the cancel button **708** based on the received information (**S616**).

[0058] In the case of being determined in **S616** that the button pressed by the user is the cancel button **708**, the CPU **401** discards the post information received in **S611** (**S617**). As a result, the post information received in **S611** is not registered in the post information database **405**, and this post information is not made public. After that, the post information registration processing ends.

[0059] In the case of being determined in **S612** that this post information does not include an image, the post information registration processing proceeds to **S618**. In addition, in the case of being determined in **S613** that the image included in the post information is not an AI-generated image, the post information registration processing proceeds to **S618**. Moreover, in the case of being determined in **S616** that the button pressed by the user is the post button **707**, the post information registration processing proceeds to **S618**.

[0060] In **S618**, the CPU **401** registers, in the post information database, the post information received in **S611**. As a result, this post information is made public, and the poster and other users who use the SNS service are able to view the registered post information, for example, post information **709** shown in FIG. **7C**. A message display section **710** of the post information **709** displays the character string that has been inputted by the poster in the character input section **702**, and a file display section **711** displays the image that has been selected by the poster in the file selection section **703**. In addition, the post information **709** may include an icon display section **712** in which the poster's icon is displayed, and a user name display section **713** in which the poster's name is displayed. When the process of **S618** is completed, the post information registration processing ends.

[0061] According to the embodiment described above, in the case of being determined that the image included in the post information accepted in **S601** is an AI-generated image, the processing

of restricting posting of this post information is performed. As a result, it is possible to suppress posting of AI-generated images that cause viewers to misidentify.

[0062] In addition, in the embodiment described above, the user I/F unit **304** displays the warning notification screen **706** that allows the poster to select whether or not to post the post information accepted in **S601**. In other words, in the case that the post information accepted in **S601** includes an AI-generated image, the poster is given an opportunity to reconsider whether or not to post this post information. As a result, it is possible to suppress posting of AI-generated images that unintentionally cause viewers to misidentify, which is performed by the poster.

[0063] In addition, in the embodiment described above, the determination of whether or not the image included in the post information is an AI-generated image is made based on whether or not the AI-generated identification information has been added to the image included in the accepted post information. As a result, in the configuration in which the AI image generation server **101** includes the AI-generated identification information adding unit **208**, it is possible to easily determine whether or not the image included in the post information is an AI-generated image.

[0064] Although the present invention has been described by using the embodiment described above, the present invention is not limited to the embodiment described above. For example, the post information may include a plurality of images. In the case that a plurality of images are included in the post information, it may be notified as to which images are AI-generated images.

[0065] It should be noted that among AI-generated images, photographs are more likely to cause to give misidentification of fictitious incidents or fictitious accidents, but illustrations are less likely to cause to give misidentification of fictitious incidents or fictitious accidents. Therefore, in the case that the AI-generated image is an illustration in **S613** described above, posting of post information including such an AI-generated image may not be restricted, and the post information registration processing may proceed to **S618**.

[0066] In addition, in the embodiment described above, the configuration, in which the determination of **S613** is made based on the presence or absence of the AI-generated identification information, has been described, but the present invention is not limited to this configuration. For example, in the determination of **S613**, an AI may be used that inputs the image included in the post information and outputs a determination result indicating whether or not the image included in the post information is an AI-generated image. It should be noted that the SNS server **103** may include a trained model for realizing this AI. Alternatively, a configuration, in which the SNS server **103** transmits the image included in the post information to an AI server including a trained model for realizing this AI and obtains a determination result from this AI server, may be adopted. By using such an AI, even in the case that the AI image generation server **101** does not include the AI-generated identification information adding unit **208**, it is possible to easily determine whether or not the image included in the post information is an AI-generated image.

[0067] In addition, in the present embodiment, the determination of **S613** may be made based on predetermined information included in the AI-generated identification information. The predetermined information is, for example, a probability or a score indicating the likelihood of an AI-generated image. For example, in **S613**, the CPU **401** converts the likelihood of an AI-generated image into a percentage, and in the case that the likelihood of a certain part being an AI-generated image is equal to or greater than a threshold value $N\%$, it is determined that the image included in the post information is an AI-generated image. On the other hand, in the case that the likelihood of a certain part being an AI-generated image is less than the threshold value $N\%$, it is determined that the image included in the post information is not an AI-generated image. It should be noted that the threshold value $N\%$ may be stored in advance in the storage unit **404** or may be included in the post information. By using such a probability or a score indicating the likelihood of an AI-generated image, it is possible to determine whether or not the image included in the post information is an AI-generated image.

[0068] In addition, in the embodiment described above, the configuration, in which as the

processing of restricting posting of the post information accepted in **S601**, the processing of displaying the warning notification screen **706**, which allows the user to select whether or not to post this post information as is, is performed, has been described, but the present invention is not limited to this configuration. For example, as the processing of restricting posting of the post information accepted in **S601**, another warning notification screen for instructing prohibition of posting of this post information may be displayed on the user I/F unit **304**. The another warning notification screen includes a message indicating that this post information cannot be posted because this post information includes an AI-generated image, and the cancel button **708**, and does not include the post button **707**. As a result, it is possible to reliably prevent posting of AI-generated images that cause viewers to misidentify.

[0069] In addition, in the embodiment described above, the configuration, in which in the case that the image included in the post information is an AI-generated image, the post information is restricted, has been described, but the present invention is not limited to this configuration. For example, in the case that the image included in the post information is an AI-generated image, a modification processing for preventing misidentification of viewers may be performed with respect to this post information.

[0070] FIG. **8A** and FIG. **8B** are flowcharts that show the steps of a second posting control processing executed by the communication system **100** shown in FIG. **1**. It should be noted that the second posting control processing is similar to the first posting control processing described above, and the following will particularly describe the differences from the first posting control processing described above. Similar to the first posting control processing described above, the second posting control processing is also performed by the communication apparatus **102** and the SNS server **103**.

[0071] FIG. **8A** is a flowchart that shows the steps of a post information transmission processing performed by the communication apparatus **102**. Similar to the post information transmission processing of FIG. **6A** described above, the post information transmission processing of FIG. **8A** is also realized by the CPU **301** of the communication apparatus **102** executing the program for the SNS application that has been stored in the SSD **303**.

[0072] As shown in FIG. **8A**, **S801** and **S802**, which are the same processes as **S601** and **S602** that have been described above, are performed, and the post information transmission processing of FIG. **8A** ends.

[0073] FIG. **8B** is a flowchart that shows the steps of a post information registration processing performed by the SNS server **103**. Similar to the post information registration processing of FIG. **6B** described above, the post information registration processing of FIG. **8B** is also realized by the CPU **401** of the SNS server **103** executing a program that has been stored in the ROM **403** or the storage unit **404**.

[0074] As shown in FIG. **8B**, **S811**, **S812**, and **S813**, which are the same processes as **S611**, **S612**, and **S613** that have been described above, are performed.

[0075] In the case of being determined in **S812** that this post information does not include an image, or in the case of being determined in **S813** that the image included in the post information is not an AI-generated image, the post information registration processing of FIG. **8B** proceeds to **S815**, which is the same process as **S618** described above. After that, the post information registration processing of FIG. **8B** ends.

[0076] In the case of being determined in **S813** that the image included in the post information is an AI-generated image, the CPU **401** performs a modification processing for preventing misidentification of viewers on this post information. Specifically, the CPU **401** adds, to an image determined to be an AI-generated image in this post information, additional information indicating that this image is an AI-generated image (**S814**). Here, the process of **S814** will be described in detail. An image **901** shown in FIG. **9A** is an example of an AI-generated image before performing the process of **S814**. In **S814**, as shown in FIG. **9B**, the CPU **401** composites additional information **902** indicating that the image **901** is an AI-generated image with the image **901**. It should be noted

that in FIG. 9B, an example is shown in which a character string “AI” has been added as the additional information **902**, but another character string or an image may be added as the additional information as long as another character string or an image is information that allows the viewer to recognize that the image is an AI-generated image. Next, the post information registration processing of FIG. 8B proceeds to **S815**, which is the same process as **S618** described above. [0077] In the embodiment described above, in the case of being determined that the image included in the accepted post information is an AI-generated image, the modification processing for preventing misidentification of viewers is performed with respect to the accepted post information. In other words, in the case that the accepted post information includes an AI-generated image, the accepted post information will not be posted at least in a state that could cause viewers to misidentify. As a result, it is possible to suppress posting of AI-generated images that cause viewers to misidentify.

[0078] In addition, in the embodiment described above, in the modification processing, the image **901** included in the accepted post information is composited with the additional information **902** indicating that the image **901** is an AI-generated image. As a result, it is possible to make a viewer who has viewed the post information including the image **901**, with which the additional information **902** has been composited, aware that there is a concern that this post information includes information that is different from the facts, thereby preventing the viewer from misidentifying.

[0079] It should be noted that in the present embodiment, as the modification processing for preventing misidentification of viewers, a processing, in which the additional information indicating that this image is an AI-generated image is added as property information of the image included in the accepted post information, may be performed. As a result, without editing the AI-generated image included in the post information, it is possible to make the viewer aware that there is a concern that this post information includes information that is different from the facts.

[0080] In addition, in the present embodiment, as the modification processing for preventing misidentification of viewers, a processing, in which a message, which indicates that an AI-generated image is included, is added to the accepted post information, may be performed.

[0081] FIG. 10A and FIG. 10B are flowcharts that show the steps of a third posting control processing executed by the communication system **100** shown in FIG. 1. It should be noted that the third posting control processing is similar to the first posting control processing and the second posting control processing that have been described above, and the following will particularly describe the differences from the first posting control processing and the second posting control processing that have been described above. Similar to the first posting control processing and the second posting control processing that have been described above, the third posting control processing is also performed by the communication apparatus **102** and the SNS server **103**.

[0082] FIG. 10A is a flowchart that shows the steps of a post information transmission processing performed by the communication apparatus **102**. Similar to the post information transmission processing of FIG. 6A and the post information transmission processing of FIG. 8A that have been described above, the post information transmission processing of FIG. 10A is also realized by the CPU **301** of the communication apparatus **102** executing the program for the SNS application that has been stored in the SSD **303**.

[0083] As shown in FIG. 10A, **S1001** and **S1002**, which are the same processes as **S601** and **S602** that have been described above, are performed, and the post information transmission processing of FIG. 10A ends.

[0084] FIG. 10B is a flowchart that shows the steps of a post information registration processing performed by the SNS server **103**. Similar to the post information registration processing of FIG. 6B and the post information registration processing of FIG. 8B that have been described above, the post information registration processing of FIG. 10B is also realized by the CPU **401** of the SNS server **103** executing a program that has been stored in the ROM **403** or the storage unit **404**.

[0085] As shown in FIG. **10B**, **S1011**, **S1012**, and **S1013**, which are the same processes as **S611**, **S612**, and **S613** that have been described above, are performed.

[0086] In the case of being determined in **S1012** that this post information does not include an image, or in the case of being determined in **S1013** that the image included in the post information is not an AI-generated image, the post information registration processing of FIG. **10B** proceeds to **S1015**, which is the same process as **S618** described above. After that, the post information registration processing of FIG. **10B** ends.

[0087] In the case of being determined in **S1013** that the image included in the post information is an AI-generated image, the CPU **401** performs a modification processing for preventing misidentification of viewers on this post information. Specifically, the CPU **401** adds, to this post information, a message indicating that an AI-generated image is included (**S1014**). Next, the post information registration processing of FIG. **10B** proceeds to **S1015**, which is the same process as **S618** described above. For example, in **S1015**, when the post information, to which the message has been added, is registered in the post information database **405**, the poster and other users who use the SNS service are able to view the registered post information, for example, post information **1101** shown in FIG. **11**. In addition to a character string **1102** inputted by the poster, the post information **1101** also includes a message **1103** indicating that an AI-generated image is included. When the process of **S1015** is completed, the post information registration processing of FIG. **10B** ends.

[0088] In the embodiment described above, as the modification processing for preventing misidentification of viewers, the processing, in which the message, which indicates that an AI-generated image is included, is added to the accepted post information, is performed. As a result, without editing the AI-generated image included in the post information, it is possible to make the viewer aware at a glance that there is a concern that this post information includes information that is different from the facts.

[0089] It should be noted that in the present embodiment, as the modification processing for preventing misidentification of viewers, a processing of excluding the AI-generated image from the post information may be performed.

[0090] FIG. **12A** and FIG. **12B** are flowcharts that show the steps of a fourth posting control processing executed by the communication system **100** shown in FIG. **1**. It should be noted that the fourth posting control processing is similar to the first posting control processing, the second posting control processing, and the third posting control processing that have been described above, and the following will particularly describe the differences from the first posting control processing, the second posting control processing, and the third posting control processing that have been described above. Similar to the first posting control processing, the second posting control processing, and the third posting control processing that have been described above, the fourth posting control processing is also performed by the communication apparatus **102** and the SNS server **103**.

[0091] FIG. **12A** is a flowchart that shows the steps of a post information transmission processing performed by the communication apparatus **102**. Similar to the post information transmission processing of FIG. **6A**, the post information transmission processing of FIG. **8A**, and the post information transmission processing of FIG. **10A** that have been described above, the post information transmission processing of FIG. **12A** is also realized by the CPU **301** of the communication apparatus **102** executing the program for the SNS application that has been stored in the SSD **303**.

[0092] As shown in FIG. **12A**, **S1201** and **S1202**, which are the same processes as **S601** and **S602** that have been described above, are performed, and the post information transmission processing of FIG. **12A** ends.

[0093] FIG. **12B** is a flowchart that shows the steps of a post information registration processing performed by the SNS server **103**. Similar to the post information registration processing of FIG.

6B, the post information registration processing of FIG. 8B, and the post information registration processing of FIG. 10B that have been described above, the post information registration processing of FIG. 12B is also realized by the CPU 401 of the SNS server 103 executing a program that has been stored in the ROM 403 or the storage unit 404.

[0094] As shown in FIG. 12B, S1211, S1212, and S1213, which are the same processes as S611, S612, and S613 that have been described above, are performed.

[0095] In the case of being determined in S1212 that this post information does not include an image, or in the case of being determined in S1213 that the image included in the post information is not an AI-generated image, the post information registration processing of FIG. 12B proceeds to S1215, which is the same process as S618 described above. After that, the post information registration processing of FIG. 12B ends.

[0096] In the case of being determined in S1213 that the image included in the post information is an AI-generated image, the CPU 401 performs a modification processing for preventing misidentification of viewers on this post information. Specifically, the CPU 401 excludes the AI-generated image from this post information (S1214). It should be noted that in the case that this post information includes a plurality of images, only AI-generated images among the plurality of images will be excluded. Next, the post information registration processing of FIG. 12B proceeds to S1215, which is the same process as S618 described above. For example, in S1215, when the post information, from which the AI-generated image has been excluded, is registered in the post information database 405, the poster and other users who use the SNS service are able to view the registered post information, for example, post information 1301 shown in FIG. 13. In the post information 1301, the placement area of the excluded AI-generated image is displayed as blank, which is shown by a reference sign 1302. When the process of S1215 is completed, the post information registration processing of FIG. 12B ends.

[0097] In the embodiment described above, as the modification processing for preventing misidentification of viewers, the processing of excluding the AI-generated image from the post information is performed. As a result, it is possible to reliably prevent misidentification of viewers that is caused by the AI-generated image.

[0098] It should be noted that in the present embodiment, in S1214, the post information may be subjected to a processing of filling in the area of the AI-generated image with a background color or another color other than the background color.

[0099] In addition, in the present embodiment, in the case that the image included in the post information is an AI-generated image, it may be possible to control whether or not to perform the modification processing on this post information based on information about the generation source of this image.

[0100] For example, image super-resolution or color conversion that uses AI generation technology generates an image based on an input image. Therefore, the generated image includes factual information, and even in the case that the post information including such a generated image is posted as is, it is unlikely to cause third parties to misidentify. On the other hand, among AI generation technologies, for example, Stable Diffusion generates images from character strings, and Deep Convolutional Generative Adversarial Networks (DCGAN) generate images from noises. There is a concern that an AI-generated image generated by the generative AI based on information other than images (non-image information) in this way is a fictitious image that is different from the facts, and when the post information including such an image is posted as is, there is a very high possibility of causing third parties to misidentify.

[0101] In order to deal with this issue, in the present embodiment, in the case that the image included in the post information is an AI-generated image, whether or not to perform the modification processing on this post information is controlled based on the information about the generation source (generation source information) of this image.

[0102] FIG. 14A and FIG. 14B are flowcharts that show the steps of a fifth posting control

processing executed by the communication system **100** shown in FIG. **1**. It should be noted that the fifth posting control processing is similar to the first posting control processing, the second posting control processing, the third posting control processing, and the fourth posting control processing that have been described above, and the following will particularly describe the differences from the first posting control processing, the second posting control processing, the third posting control processing, and the fourth posting control processing that have been described above. Similar to the first posting control processing, the second posting control processing, the third posting control processing, and the fourth posting control processing that have been described above, the fifth posting control processing is also performed by the communication apparatus **102** and the SNS server **103**.

[0103] FIG. **14A** is a flowchart that shows the steps of a post information transmission processing performed by the communication apparatus **102**. Similar to the post information transmission processing of FIG. **6A**, the post information transmission processing of FIG. **8A**, the post information transmission processing of FIG. **10A**, and the post information transmission processing of FIG. **12A** that have been described above, the post information transmission processing of FIG. **14A** is also realized by the CPU **301** of the communication apparatus **102** executing the program for the SNS application that has been stored in the SSD **303**.

[0104] As shown in FIG. **14A**, **S1401** and **S1402**, which are the same processes as **S601** and **S602** that have been described above, are performed, and the post information transmission processing of FIG. **14A** ends.

[0105] FIG. **14B** is a flowchart that shows the steps of a post information registration processing performed by the SNS server **103**. Similar to the post information registration processing of FIG. **6B**, the post information registration processing of FIG. **8B**, the post information registration processing of FIG. **10B**, and the post information registration processing of FIG. **12B** that have been described above, the post information registration processing of FIG. **14B** is also realized by the CPU **401** of the SNS server **103** executing a program that has been stored in the ROM **403** or the storage unit **404**.

[0106] As shown in FIG. **14B**, **S1411**, **S1412**, and **S1413**, which are the same processes as **S611**, **S612**, and **S613** that have been described above, are performed.

[0107] In the case of being determined in **S1412** that this post information does not include an image, or in the case of being determined in **S1413** that the image included in the post information is not an AI-generated image, the post information registration processing of FIG. **14B** proceeds to **S1416**, which is the same process as **S618** described above. After that, the post information registration processing of FIG. **14B** ends.

[0108] In the case of being determined in **S1413** that the image included in the post information is an AI-generated image, the CPU **401** determines whether or not the generation source of this image is the information other than images (the non-image information) based on the generation source information included in the AI-generated identification information of this image (**S1414**).

[0109] In the case of being determined in **S1414** that the generation source of this image is not the information other than images (the non-image information), that is, in the case that the generation source of this image is an image, the post information registration processing of FIG. **14B** proceeds to **S1416**. Here, in the case that the generation source of this image is an image, since this image is an image generated by the generative AI based on an input image, that is, this image is an image that includes factual information, even in the case that the post information including such an image is posted as is, it is unlikely to cause third parties to misidentify. Therefore, in **S1416**, the CPU **401** registers the post information received in **S1411** in the post information database **405** as is. After that, the post information registration processing of FIG. **14B** ends.

[0110] In the case of being determined in **S1414** that the generation source of this image is the information other than images (the non-image information), the post information registration processing of FIG. **14B** proceeds to **S1415**. Here, in the case that the generation source of this

image is the information other than images (the non-image information), this image is an image generated by the generative AI based on the information other than images (the non-image information), such as character strings or noises. There is a concern that an AI-generated image generated by the generative AI based on information other than images (non-image information) in this way is a fabricated image of a fictitious incident or a fictitious accident that is different from the facts, and when the post information including such an image is posted as is, there is a very high possibility of causing third parties to misidentify. Therefore, in **S1415**, the CPU **401** performs a modification processing for preventing misidentification of viewers on the post information received in **S1411**. It should be noted that the modification processing is, for example, as described above, a processing of compositing additional information indicating that the image is an AI-generated image with the AI-generated image. In addition, the modification processing is a processing of adding additional information indicating that the image is an AI-generated image as property information of the AI-generated image. In addition, the modification processing is a processing of adding a message indicating that an AI-generated image is included. In addition, the modification processing is a processing of excluding the AI-generated image from the post information. Next, the post information registration processing of FIG. **14B** proceeds to **S1416**.

[0111] In the embodiment described above, in the case of being determined that the image included in the accepted post information is an image generated by the generative AI using an input image as input, the modification processing for preventing misidentification of viewers will not be performed. On the other hand, in the case of being determined that the image included in the accepted post information is an image generated by the generative AI using information other than the input image as input, the modification processing for preventing misidentification of viewers will be performed. As a result, it is possible to perform the above-described modification processing on the AI-generated image that is likely to include information that is different from the facts, while preventing the modification processing from being excessively performed on the AI-generated image that does not require modification.

[0112] It should be noted that in the embodiment described above, the configuration, in which the SNS server **103** determines whether or not the image included in the post information is an AI-generated image, has been described, but the present invention is not limited to this configuration. For example, the communication apparatus **102** on which the SNS application has been installed may determine whether or not the image included in the post information is an AI-generated image.

[0113] FIG. **15** is a flowchart that shows the procedure of a post information transmission processing executed by the communication apparatus **102** shown in FIG. **1**. It should be noted that the post information transmission processing of FIG. **15** is similar to the first posting control processing described above, and the following will particularly describe the differences from the first posting control processing described above. Similar to the post information transmission processing of FIG. **6A** described above, the post information transmission processing of FIG. **15** is also realized by the CPU **301** of the communication apparatus **102** executing the program for the SNS application that has been stored in the SSD **303**.

[0114] As shown in FIG. **15**, the CPU **301** performs **S1501**, which is the same process as **S601** described above. Next, the CPU **301** analyzes the accepted post information and determines whether or not this post information includes an image (**S1502**).

[0115] In the case of being determined in **S1502** that this post information includes an image, the CPU **401** determines whether or not the image included in the post information is an AI-generated image (**S1503**). It should be noted that in the present embodiment, as in **S613** described above, the determination of **S1503** may be made based on the presence or absence of the AI-generated identification information. In addition, the determination of **S1503** may be made by using an AI that inputs the image included in the post information and outputs a determination result indicating whether or not the image included in the post information is an AI-generated image. It should be noted that in the present embodiment, the communication apparatus **102** may include a trained

model for realizing this AI. Alternatively, a configuration, in which the communication apparatus **102** transmits the image included in the post information to an AI server including a trained model for realizing this AI and obtains a determination result from this AI server, may be adopted.

[0116] In the case of being determined in **S1503** that the image included in the post information is an AI-generated image, the CPU **301** performs a processing of restricting posting of the post information accepted in **S1501**. Specifically, the CPU **301** causes the warning notification screen **706** to be displayed on the posting screen **701** displayed on the user I/F unit **304**.

[0117] Next, the CPU **301** accepts a button operation such as pressing either the post button **707** or the cancel button **708** on the warning notification screen **706** by the user (**S1505**). Next, the CPU **301** determines whether the button pressed by the user is the post button **707** or the cancel button **708** (**S1506**).

[0118] In the case of being determined in **S1506** that the button pressed by the user is the cancel button **708**, the CPU **301** discards the post information accepted in **S1501** (**S1507**). After that, the post information transmission processing of FIG. **15** ends.

[0119] In the case of being determined in **S1502** that this post information does not include an image, the post information transmission processing of FIG. **15** proceeds to **S1508**. In addition, also in the case of being determined in **S1503** that the image included in the post information is not an AI-generated image, the post information transmission processing of FIG. **15** proceeds to **S1508**. Furthermore, also in the case of being determined in **S1506** that the button pressed by the user is the post button **707**, the post information transmission processing of FIG. **15** proceeds to **S1508**.

[0120] In **S1508**, the CPU **301** registers, in the post information database, the post information accepted in **S1501**. After that, the post information transmission processing of FIG. **15** ends.

[0121] In the embodiment described above, in the communication apparatus **102** on which the SNS application has been installed, in the case of being determined that the image included in the accepted post information is an AI-generated image, the processing of restricting posting of this post information is performed. As a result, it is possible to suppress posting of AI-generated images that cause viewers to misidentify.

[0122] It should be noted that also in the above-described post information transmission processing of FIG. **15**, similar to the above-described configuration, in the case that the image determined to be an AI-generated image in **S1503** is an illustration, posting of the post information including this AI-generated image may not be restricted and the post information transmission processing of FIG. **15** may proceed to **S1508**.

[0123] In addition, also in the above-described post information transmission processing of FIG. **15**, similar to the above-described configuration, the determination of **S1503** may be made based on predetermined information included in the AI-generated identification information, for example, a probability or a score indicating the likelihood of an AI-generated image.

[0124] In addition, also in the above-described post information transmission processing of FIG. **15**, similar to the above-described configuration, in the case that the image included in the post information is an AI-generated image, control to prohibit posting of this post information may be performed. Specifically, in **S1504**, another warning notification screen, which includes a message indicating that this post information cannot be posted because this post information includes an AI-generated image, and the cancel button **708**, and does not include the post button **707**, is displayed on the user I/F unit **304**. When the user presses the cancel button **708**, the post information accepted in **S1501** is discarded (**S1507**). As a result, it is possible to reliably prevent posting of AI-generated images that cause viewers to misidentify.

[0125] In addition, in the present embodiment, in the case that the image included in the post information is an AI-generated image, the communication apparatus **102**, on which the SNS application has been installed, may perform a modification processing for preventing misidentification of viewers with respect to this post information.

[0126] FIG. **16** is a flowchart that shows another procedure of the post information transmission

processing executed by the communication apparatus **102** shown in FIG. **1**. It should be noted that the post information transmission processing of FIG. **16** is similar to the post information transmission processing of FIG. **15** that has been described above, and the following will particularly describe the differences from the post information transmission processing of FIG. **15** that has been described above. Similar to the post information transmission processing of FIG. **15** that has been described above, the post information transmission processing of FIG. **16** is also realized by the CPU **301** of the communication apparatus **102** executing the program for the SNS application that has been stored in the SSD **303**.

[0127] As shown in FIG. **16**, the CPU **301** performs **S1601**, **S1602**, and **S1603**, which are the same processes as **S1501**, **S1502**, and **S1503** that have been described above.

[0128] In the case of being determined in **S1603** that the image included in the post information is an AI-generated image, the CPU **301** performs a modification processing for preventing misidentification of viewers on the post information accepted in **S1601**. It should be noted that the modification processing is, for example, as described above, a processing of compositing additional information indicating that the image is an AI-generated image with the AI-generated image. In addition, the modification processing is a processing of adding additional information indicating that the image is an AI-generated image as property information of the AI-generated image. In addition, the modification processing is a processing of adding a message indicating that an AI-generated image is included. In addition, the modification processing is a processing of excluding the AI-generated image from the post information. After that, the CPU **301** registers, in the post information database **405**, the post information on which the modification processing has been performed (**S1605**), and the post information transmission processing of FIG. **16** ends.

[0129] In the case of being determined in **S1602** that this post information does not include an image, the post information transmission processing of FIG. **16** proceeds to **S1605**. In addition, also in the case of being determined in **S1603** that the image included in the post information is not an AI-generated image, the post information transmission processing of FIG. **16** proceeds to **S1605**. In **S1605**, the CPU **301** registers, in the post information database **405**, the post information accepted in **S1601** as is. After that, the post information transmission processing of FIG. **16** ends.

[0130] In the embodiment described above, in the communication apparatus **102** on which the SNS application has been installed, in the case of being determined that the image included in the accepted post information is an AI-generated image, the modification processing for preventing misidentification of viewers is performed with respect to the accepted post information. In other words, in the case that the accepted post information includes an AI-generated image, the accepted post information will not be posted at least in a state that could cause viewers to misidentify. As a result, it is possible to suppress posting of AI-generated images that cause viewers to misidentify.

[0131] It should be noted that in the post information transmission processing of FIG. **16** that has been described above, in the case of being determined in **S1603** that the image included in the post information is an AI-generated image, the CPU **301** may perform the same determination processing as in **S1414** described above. In the case of being determined in this determination that the generation source of this image is not the information other than images (the non-image information), that is, in the case that the generation source of this image is an image, the post information transmission processing of FIG. **16** proceeds to **S1605**. On the other hand, in the case of being determined in this determination that the generation source of this image is the information other than images (the non-image information), the post information transmission processing of FIG. **16** proceeds to **S1604**. By controlling in this way, it is possible to perform the above-described modification processing on the AI-generated image that is likely to include information that is different from the facts, while preventing the modification processing from being excessively performed on the AI-generated image that does not require modification.

[0132] It should be noted that in the present embodiment, in the case that the post information includes an AI-generated image, the configuration that restricts posting of the post information, the

configuration that prohibits posting of the post information, and the configuration that performs the modification processing on the post information have been described, but the present invention is not limited to these configurations. For example, in the case that post information includes audio data generated by the generative AI, posting of the post information may be restricted, or posting of the post information may be prohibited, or a modification processing may be performed on the post information. For example, this modification processing is a processing of adding information indicating that the audio data has been generated by the generative AI as property information of the audio data, or a processing of adding, to the post information, a message indicating that the audio data generated by the generative AI is included.

Other Embodiments

[0133] Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

[0134] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0135] This application claims the benefit of Japanese Patent Application No. 2024-025433, filed on Feb. 22, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. A non-transitory computer-readable storage medium storing an application program for using a service that publishes post information for causing an information processing apparatus comprising the application program to execute: a determining step of determining whether or not an image included in accepted post information is an image generated by a generative AI; and a restricting step of, in a case of being determined that the image included in the accepted post information is an image generated by the generative AI, restricting posting of the accepted post information.
2. The non-transitory computer-readable storage medium according to claim 1, wherein in the restricting step, a screen for allowing a poster of the accepted post information to select whether or not to post the accepted post information is displayed on a display unit of the information processing apparatus.
3. The non-transitory computer-readable storage medium according to claim 1, wherein in the restricting step, a screen for allowing a poster of the accepted post information to instruct cancellation of posting of the accepted post information is displayed on a display unit of the

information processing apparatus.

- 4.** A non-transitory computer-readable storage medium storing an application program of an application for using a service that publishes post information for causing an information processing apparatus comprising the application program to execute: a determining step of determining whether or not an image included in accepted post information is an image generated by a generative AI; and a step of, in a case of being determined that the image included in the accepted post information is an image generated by the generative AI, performing a modification processing for preventing misidentification of viewers on the accepted post information.
- 5.** The non-transitory computer-readable storage medium according to claim 4, wherein in the modification processing, additional information indicating that the image included in the accepted post information is an image generated by the generative AI is composited with the image included in the accepted post information.
- 6.** The non-transitory computer-readable storage medium according to claim 4, wherein in the modification processing, additional information indicating that the image included in the accepted post information is an image generated by the generative AI is added as property information of the image included in the accepted post information.
- 7.** The non-transitory computer-readable storage medium according to claim 4, wherein in the modification processing, a message indicating that an image generated by the generative AI is included is added to the accepted post information.
- 8.** The non-transitory computer-readable storage medium according to claim 4, wherein in the modification processing, an image generated by the generative AI is excluded from the accepted post information.
- 9.** The non-transitory computer-readable storage medium according to claim 4, wherein the information processing apparatus comprising the application program is caused to, in a case of being determined that the image included in the accepted post information is an image generated by the generative AI using an input image as input, not execute the step of performing the modification processing, and in a case of being determined that the image included in the accepted post information is an image generated by the generative AI using information other than the input image as input, execute the step of performing the modification processing.
- 10.** The non-transitory computer-readable storage medium according to claim 1, wherein identification information indicating having been generated by the generative AI is added to an image generated by the generative AI, and the determining step is performed based on whether or not the identification information has been added to the image included in the accepted post information.
- 11.** The non-transitory computer-readable storage medium according to claim 4, wherein identification information indicating having been generated by the generative AI is added to an image generated by the generative AI, and the determining step is performed based on whether or not the identification information has been added to the image included in the accepted post information.
- 12.** The non-transitory computer-readable storage medium according to claim 1, wherein the determining step is performed by using an AI that inputs the image included in the accepted post information and outputs a determination result indicating whether or not to have been generated by the generative AI.
- 13.** The non-transitory computer-readable storage medium according to claim 4, wherein the determining step is performed by using an AI that inputs the image included in the accepted post information and outputs a determination result indicating whether or not to have been generated by the generative AI.
- 14.** A control method for an information processing apparatus comprising an application program for using a service that publishes post information, the control method comprising: a step of determining whether or not an image included in accepted post information is an image generated

by a generative AI; and a step of, in a case of being determined that the image included in the accepted post information is an image generated by the generative AI, restricting posting of the accepted post information.

15. A control method for an information processing apparatus comprising an application program for using a service that publishes post information, the control method comprising: a step of determining whether or not an image included in accepted post information is an image generated by a generative AI; and a step of, in a case of being determined that the image included in the accepted post information is an image generated by the generative AI, performing a modification processing for preventing misidentification of viewers on the accepted post information.

16. An information processing apparatus comprising an application program for using a service that publishes post information, the information processing apparatus comprising: at least one processor; and a memory coupled to the processor storing instructions that, when executed by the processor, cause the processor to function as: a determining unit that determines whether or not an image included in accepted post information is an image generated by a generative AI; and a restricting unit that, in a case of being determined that the image included in the accepted post information is an image generated by the generative AI, restricts posting of the accepted post information.

17. An information processing apparatus comprising an application program for using a service that publishes post information, the information processing apparatus comprising: at least one processor; and a memory coupled to the processor storing instructions that, when executed by the processor, cause the processor to function as: a determining unit that determines whether or not an image included in accepted post information is an image generated by a generative AI; and a processing unit that, in a case of being determined that the image included in the accepted post information is an image generated by the generative AI, performs a modification processing for preventing misidentification of viewers on the accepted post information.
